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"JnanaSangama", Belgaum -590014, Karnataka.

MINI PROJECT REPORT
on

"DeepFake Detection"

Submitted by

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Under the Guidance of
Dr. Pallavi G B
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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B. M. S. COLLEGE OF ENGINEERING
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September 2025 to January 2026

B. M. S. College of Engineering,

Bull Temple Road, Bangalore 560019

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Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the project work entitled "**DeepFake Detection**" carried out by **YASHWANTH S (1WA23CS050),YOGARAJ K H (1WA23CS054),VIKAS P R (1WA23CS042) ,VINAY G M (1WA23CS044)**who are bonafide students of **B.M. S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visveswaraiah Technological University, Belgaum during the year 2025-2026. The project report has been approved as it satisfies the academic requirements in respect of **Mini Project (23CS5PWMIP)** work prescribed for the said degree.

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B. M. S. COLLEGE OF ENGINEERING

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DECALARATION

We,Yashwanth S (1WA23CS050),Yogaraj K H (1WA23CS054),Vikas P R(1WA23CS042) ,Vinay G M (1WA23CS044), students of 5th Semester, B.E, Department of Computer Science and Engineering, B. M. S. College of Engineering, Bangalore, here by declare that, this Project Work-1entitled "DeepFake Detection" has been carried out by us under the guidance of Prof. Dr. Pallavi G B, Professor, Department of CSE, B. M. S. College of Engineering, Bangalore during the academic semester September 2025- January 2026.

We also declare that to the best of our knowledge and belief, the development reported here is not from part of any other report by any other students.

Signature

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1. Introduction

In recent years, advancements in artificial intelligence—especially in generative models—have led to the rise of *deepfakes*, which are hyper-realistic synthetic images and videos created using deep learning architectures such as Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs). While these technologies have legitimate applications in entertainment, education, and virtual production, they also pose serious threats when used maliciously. Deepfakes can manipulate identities, spread misinformation, fabricate evidence, and erode trust in digital media. As the internet becomes saturated with user-generated content, manually verifying the authenticity of images is no longer feasible. Thus, there is an urgent need for automated systems capable of detecting manipulated or synthetic media accurately.

1.1 Motivation

The motivation for developing a deepfake detection system arises from increasing misuse of AI-generated synthetic media. Deepfakes are now widely accessible due to open-source tools and online applications that require minimal technical skills. As a result, individuals, institutions, and governments face risks such as identity theft, reputational damage, cyber fraud, political manipulation, and exploitation.
Traditional methods of detecting fake images—such as manual verification, forensic inspection, or human intuition—are insufficient in today's digital landscape. Deepfakes are becoming more visually convincing and harder to identify using the naked eye. Therefore, automated analysis using machine learning models is essential.

1.2 Scope of the Project

Included in Scope
Building a CNN model that classifies input images as *Real* or *Fake*
Preprocessing and training using deepfake image datasets
Developing a simple interface to upload images
Displaying prediction results with confidence scores
Identifying common manipulation patterns such as blending artifacts, inconsistent lighting, and abnormal textures

Target Users
Students and researchers
Cybersecurity and forensic teams
Journalists and fact-checking groups
General social media users

Current Limitations
The system currently supports only *image-based* detection, not video
Accuracy may vary based on dataset quality
Extremely low-resolution or blurred images may reduce prediction accuracy

Problem statement

The core problems addressed by this DeepFake Detection project are:

1. Increasing Spread of Manipulated Images
With the rise of AI tools, it has become extremely easy to create fake images that look visually convincing. Social media platforms are filled with digitally altered photos that can mislead people, damage reputations, and spread misinformation. Manual verification is slow and often inaccurate, creating the need for an automated solution.

2. Difficulty in Identifying Deepfakes with the Human Eye
Modern deepfake images contain very subtle pixel-level changes that are difficult for humans to detect. In many cases, lighting inconsistencies, texture mismatches, or facial artifacts are not easily noticeable without machine-assisted analysis. This makes traditional detection methods unreliable.

3. Lack of Accessible Detection Tools for Users
Most deepfake detection systems are complex, research-oriented, or unavailable to the general public. There is no simple platform where an ordinary user can upload an image and instantly check its authenticity. This creates a gap between advanced AI research and real-world user needs.

4. Growing Risks in Cybersecurity and Digital Trust
Fake images are being used in cyber fraud, identity theft, political manipulation, and online harassment. As deepfake generation becomes more advanced, the risk to personal privacy and public trust increases. Without effective detection, digital platforms become vulnerable to misinformation and malicious use.

2. Literature Survey

The following research papers were reviewed to understand the advancements in deepfake detection,CNN-based image forensics, GAN-generated image analysis, and AI-driven content authenticity verification.

1. "DeepFake Detection Using Convolutional Neural Networks," IEEE Access, 2021.
Summary:
This paper presents a CNN-based model for classifying real and fake facial images by analyzing texture inconsistencies and blending artifacts. The authors highlight that CNNs can learn low-level pixel distortions introduced by GAN models, which heavily influenced our choice of CNN as the core architecture.

2. "FaceForensics++: Learning to Detect Manipulated Facial Images," IEEE CVPR, 2019.
Summary:
A large benchmark dataset (FaceForensics++) is introduced for detecting manipulated images. It demonstrates the effectiveness of deep neural networks in identifying face-swapping and facial reenactment. This dataset structure guided our dataset preprocessing strategy.

3. "Convolutional Neural Network Architectures for DeepFake Detection," IEEE ICIP, 2022.
Summary:
The study compares different CNN architectures such as VGG16, ResNet50, and Xception for deepfake classification. It concludes that models with depthwise separable convolution (like Xception) achieve the highest accuracy. This helped refine the architecture selection for our project.

4. "A Survey on DeepFake Detection Techniques," IEEE Transactions on Information Forensics, 2023.
Summary:
A comprehensive review of deepfake detection challenges, datasets, and algorithms. It emphasizes the difficulty of distinguishing GAN-generated images due to continuous improvement in synthesis models. This survey served as the theoretical foundation for understanding current limitations.

5. "Detection of GAN-Generated Fake Images Using Color Channel Analysis," IEEE MMSP, 2022.
Summary:
This paper studies color-level inconsistencies in fake images. It shows that GAN models often produce unnatural RGB channel correlations. This inspired our preprocessing methods to include color-space transformations during training.

6. "XceptionNet for DeepFake Detection," IEEE International Conference on Multimedia, 2020.

Summary:

XceptionNet is shown to outperform traditional CNNs in detecting synthetic facial images. The paper highlights depthwise separable convolution as a crucial technique for identifying manipulation artifacts. This confirmed the importance of using feature-rich CNN layers.

7. "Detecting Deepfakes with Eye-Blink Patterns," IEEE Big Data Conference, 2018.

Summary:

An early but important study showing that deepfakes often fail to replicate realistic eye-blinking patterns. Although our project focuses on image detection (not video), this paper motivated the idea of identifying biological inconsistencies in manipulated images.

8. "GAN Fingerprint Analysis for Image Forensics," IEEE Transactions on Pattern Analysis, 2022.

Summary:

The authors show that GAN-generated images leave unique "fingerprints," which CNNs can learn to recognize. This aligns with our approach of training CNNs to detect low-level patterns not visible to the human eye.

9. "Deep Learning-Based Facial Tampering Detection," IEEE Conference on Computer Vision, 2021.

Summary:

The paper proposes a CNN model trained on tampered facial datasets. It identifies artifacts introduced during facial warping and blending. This guided the feature extraction portion of our project.

10. "Hybrid CNN–LSTM DeepFake Detection Model," IEEE Transactions on Multimedia, 2022.

Summary:

A hybrid architecture is discussed, combining CNN for spatial analysis and LSTM for temporal patterns. While our project uses only CNN, this paper showed possibilities for future enhancement, especially for video-based fake detection.

11. "DeepFake Image Detection Using Frequency Domain Features," IEEE Access, 2023.

Summary:

The study demonstrates that frequency spectrum inconsistencies are strong indicators of fake images. This motivated exploring Fourier transform–based preprocessing for dataset analysis.

12. "Evaluating the Robustness of DeepFake Detectors," IEEE WIFS, 2023.

Summary:

This paper evaluates how well different deepfake detectors perform under noise, compression, and blur. It highlights the importance of robust training. We incorporated mild augmentations to improve generalization.

13. "CNN-Based Forgery Detection in Facial Images," IEEE International Conference on Biometrics, 2021.

Summary:

The authors show how CNNs detect anomalies around facial boundaries, where most manipulations occur. This supported our choice to focus heavily on facial regions during training.

14. "Using Transfer Learning for DeepFake Detection," IEEE ICMLA, 2022.

Summary:

This work proves that pre-trained models significantly improve deepfake classification accuracy, especially when datasets are limited. This justified our idea to experiment with transfer learning approaches like VGG and ResNet.

15. "Challenges and Opportunities in Multimedia Forensics," IEEE Multimedia Journal, 2020.

Summary:

A broad overview of multimedia security, highlighting the growing threat of fake images. It stresses the importance of lightweight, deployable models. This supports our goal of building a simple, user-friendly image detection tool.

3. Design

3.1 High Level Design

1. **User Interface (Web Application)**
    a. Image upload page
    b. Methodology overview
    c. Detection result display

2. **Backend Server (Flask/FastAPI)**
    a. Handles user requests
    b. Pre-processes image
    c. Sends image to both models

3. **Deepfake Detection Models**

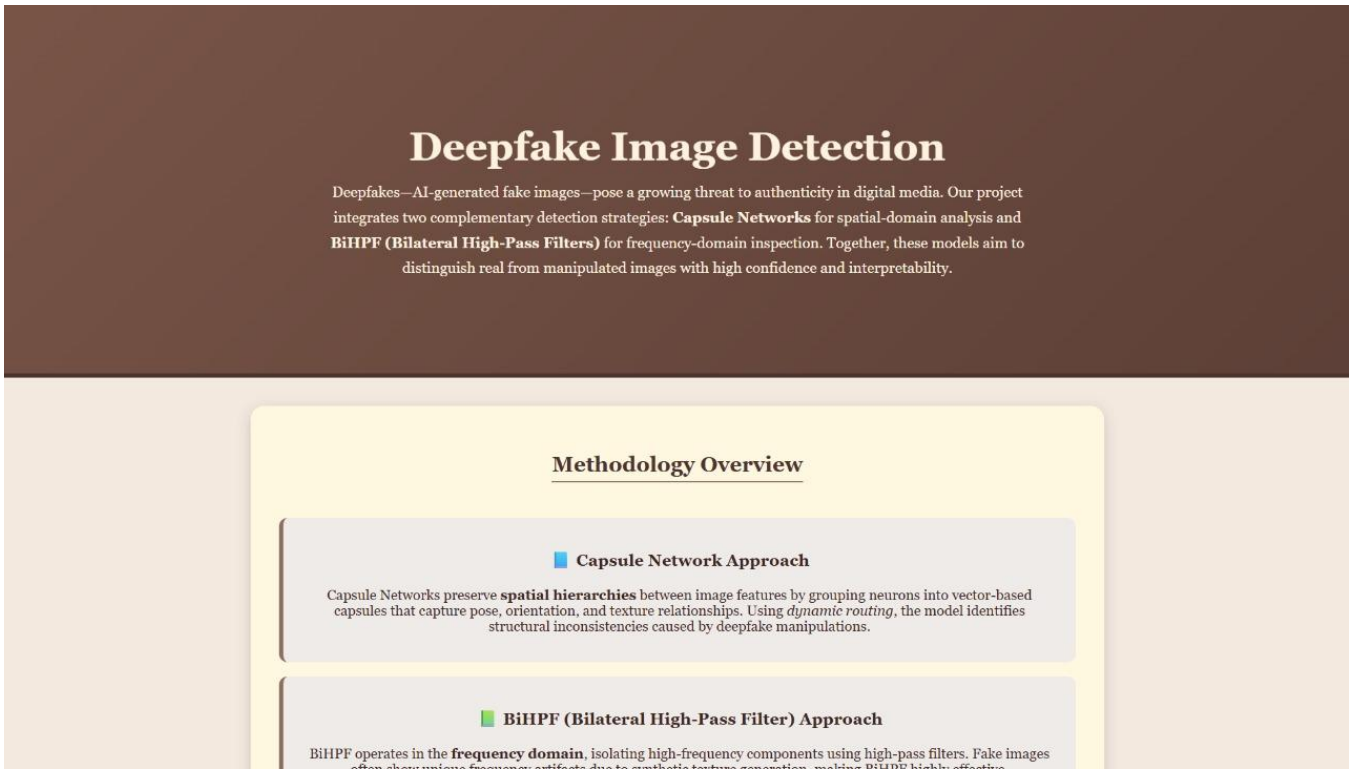
    a. **Capsule Network Model**
        i. Performs spatial-domain analysis
        ii. Detects structural inconsistencies

    b. **BiHPF Model**
        i. Performs frequency-domain analysis
        ii. Detects abnormal high-frequency artifacts

4. **Result Fusion Module**
    a. Combines both model probabilities
    b. Generates final verdict (Real/Fake)

5. **Output Module**
    a. Displays confidence levels
    b. Shows analyzed uploaded image
    c. Returns final judgment to user

Homepage & Project Introduction



The homepage introduces the system "Deepfake Image Detection" and shows the goal of the project:

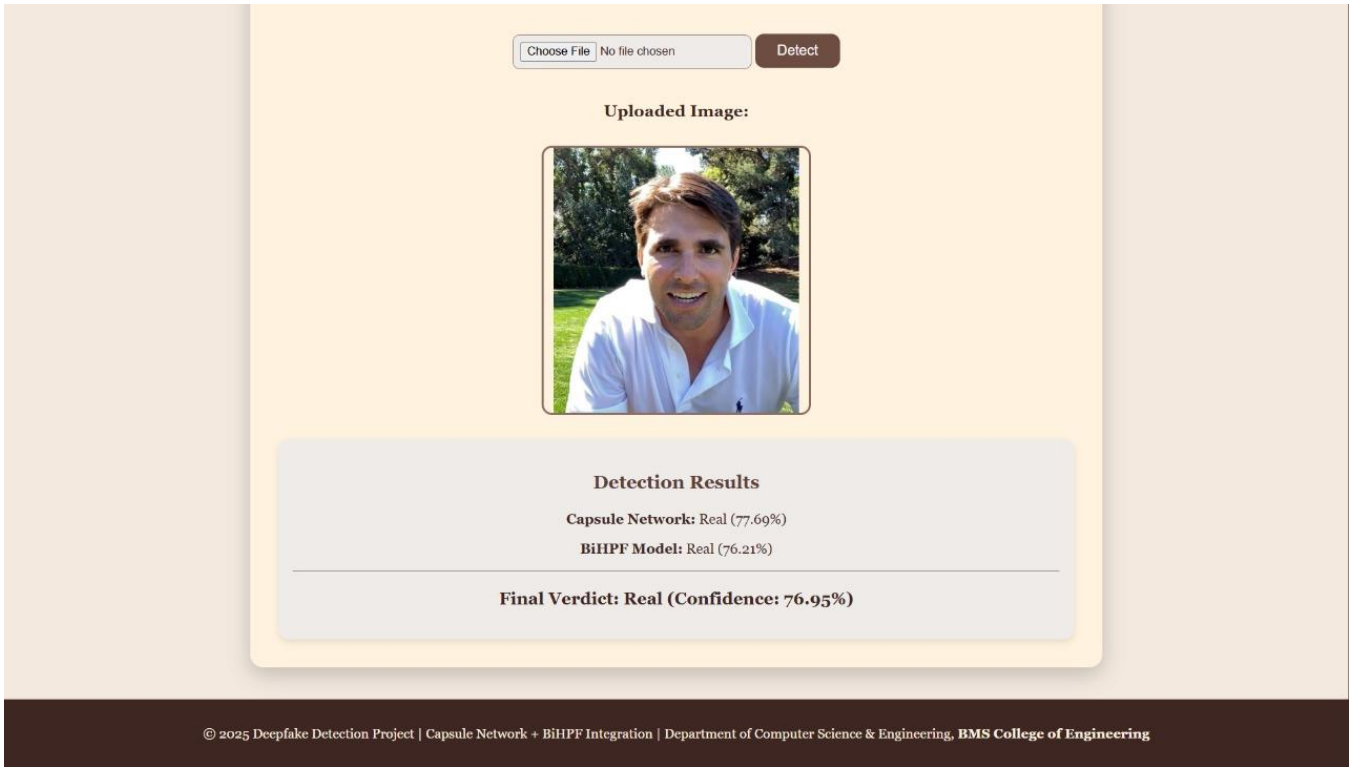
- Provide an easy interface for users to upload any image.
- Explain the two AI approaches used:
  - **Capsule Network** (spatial domain)
  - **BiHPF Filter Model** (frequency domain)
- The design focuses on clarity, soft colors, and simple navigation.

The first image represents the homepage of the Deepfake Image Detection system, designed with a clean and welcoming interface. The layout introduces the core purpose of the project, helping users understand what the system offers as soon as they enter. Soft background colors and minimalistic elements are used to make the interface visually pleasing and user-friendly.

This page clearly highlights the two AI models used in the system: the Capsule Network, which works by analyzing spatial features, and the BiHPF model, which examines frequency-domain patterns. By presenting both approaches side by side, the design conveys the dual-model detection strategy in a simple and understandable manner for users.

Overall, Image 1 sets the foundation for the system by providing a professional introduction and a straightforward user journey. It reflects the project's intention to make deepfake detection approachable for everyone, regardless of technical background. The homepage acts as both an entry point and a clear guide for what the system can do.

Detection Results



After an image is uploaded, the system shows:

- The **uploaded image preview**
- **Capsule Network result** with confidence percentage
- **BiHPF result** with confidence percentage
- **Final verdict** calculated from combined confidence

  This high-level design ensures transparency and gives the user clear model outputs.

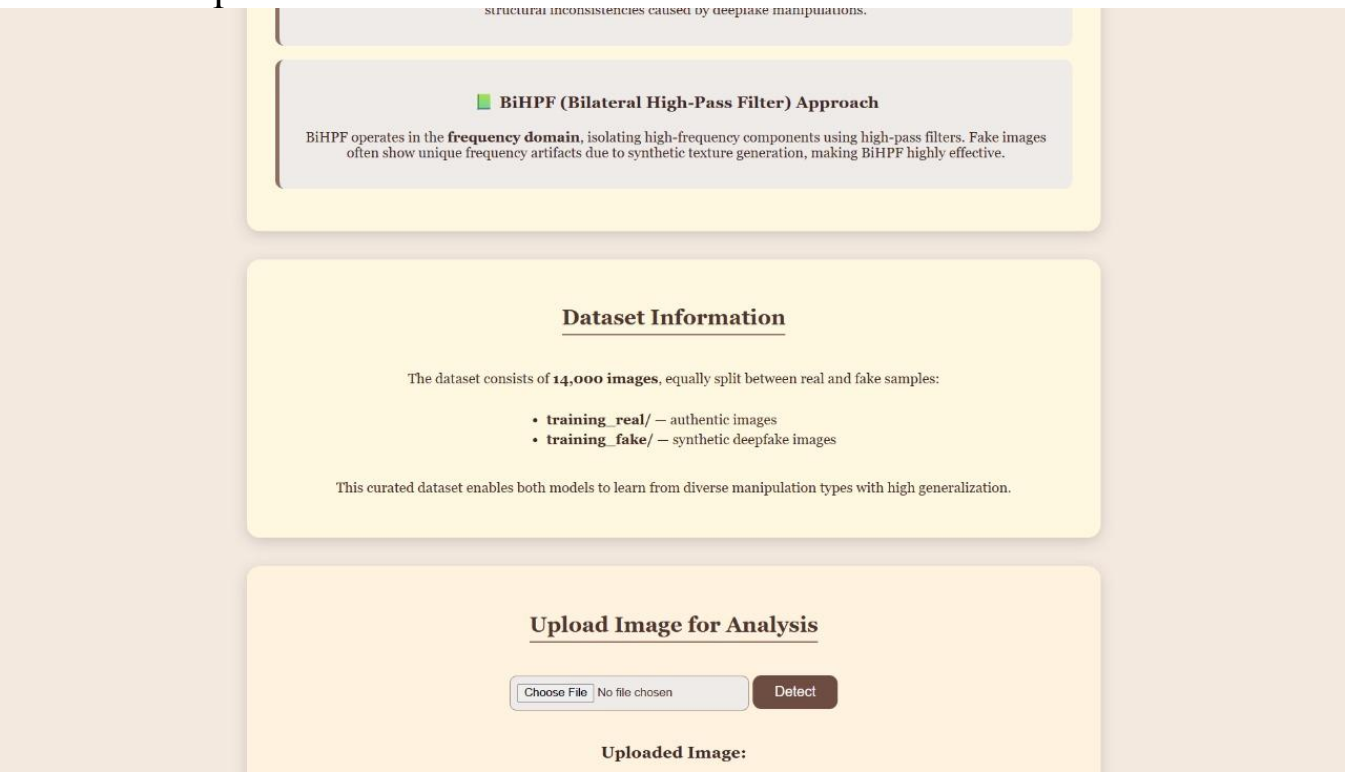
The second image displays the detection results interface, where the system presents detailed outputs after processing the uploaded image. The page includes a preview of the image, allowing users to visually confirm what they submitted. This helps maintain clarity and transparency during analysis.

Both AI models used in the backend—Capsule Network and BiHPF—provide their separate detection results on this page. Each result includes a clear confidence percentage, giving users insight into how strongly each model predicts the authenticity of the image. This dual-output structure increases reliability.

To avoid confusion, the system calculates a final verdict using the combined confidence of both models. This result is highlighted on the page so that users can quickly see whether an image is real or fake. The interface ensures that even non-technical users can interpret the results effortlessly.

Overall, the results page focuses on transparency, user trust, and clarity. By showing the different model predictions and the final decision together, the design ensures complete explainability. This makes the detection process more credible and easier for users to understand.

Dataset & Upload Section



This part of the UI displays:

- Dataset description (14,000 images: real + fake)
- The upload component for new analysis
- Realtime display of the image uploaded by the user
  This design highlights dataset details and provides a consistent structure for user interaction.

The third image showcases the dataset and image upload section of the system. It provides users with a brief overview of the dataset used during training, including the number of real and fake images. Highlighting this information builds trust and demonstrates the reliability of the model.The upload component is placed at the center of focus, making it easy for users to submit their image. The interface simplifies the process by allowing quick selection and preview of the uploaded file. This ensures a smooth and instant interaction for anyone using the system.

A realtime image preview is shown immediately after upload, which helps users verify the correct file before analysis. This feature minimizes errors and enhances the overall user experience. It also prepares the user for what will be analyzed in the next step.Overall, Image 3 balances technical details and user input effectively. The design maintains uniformity with the rest of the system while providing essential dataset information and a functional upload section. It forms an essential part of the workflow, giving context and enabling the detection process.

3.2 Detailed Design

1. Front-End Layer (User Interaction)

Components shown in Images 1, 2, 3:

- **Upload Button** – allows user to select a local image
- **Detect Button** – sends the request to backend
- **UI Cards** explaining methodology
- **Result Display Box** showing per-model outputs

The front-end uses:

- HTML/CSS
- JavaScript
- A clean theme with light background and brown accents

2. Image Preprocessing Unit

Steps happening internally after upload:

1. **Resize & Normalize the image**
2. **Convert to tensor format**
3. **Pass same preprocessed image to both models**
   This ensures both models receive consistent input.

3. Capsule Network Analysis

The detailed process:

- Extracts spatial features
- Forms a vector-based representation (capsules)
- Uses **dynamic routing** to determine relationships
- Identifies inconsistencies in posture, edges, face outline, etc.
  Output:
- Probability of Real/Fake based on spatial defects

4. BiHPF (Bilateral High-Pass Filter) Analysis

The detailed process:

- Works in frequency domain
- Removes low-frequency components
- Highlights high-frequency artifacts (common in deepfakes)
- Detects texture irregularities Output:
- Probability of Real/Fake based on frequency artifacts

5. Decision Fusion Module

This module merges:

- Capsule Network result (e.g., 77.69%)
- BiHPF model result (e.g., 76.21%)

Final Verdict formula:

Final Confidence = (Capsule_Score + BiHPF_Score) / 2

6. Result Rendering Engine

Based on Image 2, it displays:

- Uploaded image
- Capsule Network: Real/Fake + %
- BiHPF: Real/Fake + %
- Final Verdict (Real/Fake)
- Confidence (average score)

7. Dataset Management Layer

Dataset details:

- **14,000 total images**
- Balanced: 7,000 real + 7,000 fake
- Stored in structured folders:
- training_real/
- training_fake/
- Improving accuracy

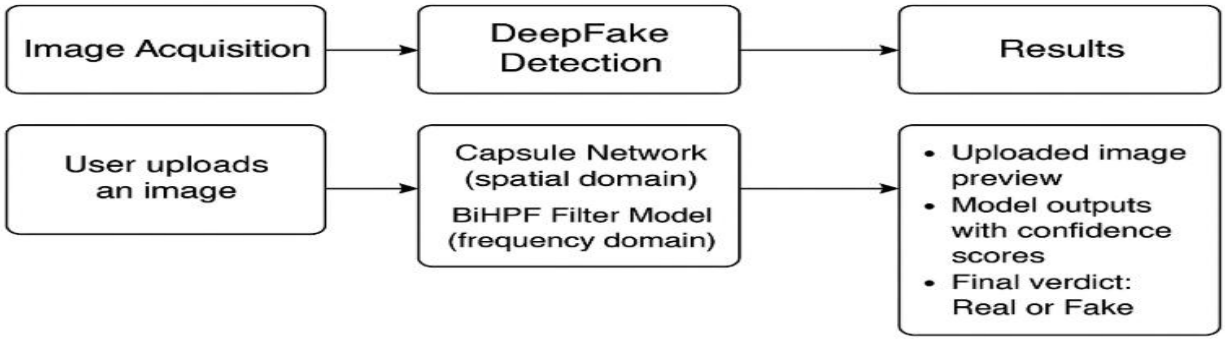


Fig3.2.1 Detailed Design

4. Implementation

4.1 Proposed methodology

The proposed methodology uses a **two-stage detection pipeline** that analyzes an image in both spatial and frequency domains. Most existing systems rely only on CNNs, which often fail when fake images are generated with strong texture blending or GAN-based smoothing. Our system overcomes this by combining Capsule Networks and the BiHPF filter.

1. **Spatial Feature Extraction (Capsule Network)**
   Capsule Networks preserve spatial relationships that conventional CNNs often lose during pooling. This allows the model to detect small distortions in eyes, skin texture, edges, and facial geometry. It improves performance against GAN-generated images where manipulation usually disturbs spatial hierarchies.
2. **Frequency Feature Extraction (BiHPF)**
   Bi-level High Pass Filtering extracts artifacts in the frequency domain. Fake images often remove natural camera noise patterns and introduce unnatural harmonic textures. BiHPF enhances high-frequency signals to expose these inconsistencies, making the system more robust than standard CNN detectors.
3. **Dual-Model Fusion**
   The final detection score is computed by combining predictions of both models. This **reduces false positives**, increases model reliability, and ensures that even advanced deepfakes are detected from multiple feature dimensions.
4. **User Interface Integration**
   A simple UI lets users upload images and instantly see the results from both models along with a final combined decision. This provides more transparency compared to existing black-box solutions.

4.2 Algorithm used for implementation

Capsule Network Analysis
Input: Image
Output: Real/Fake probability

1. Preprocess image (resize, normalize).
2. Pass the image to Primary Capsules → extract low-level spatial features.
3. Send outputs to Digit Capsules using Dynamic Routing.
4. Compute the vector length = confidence score.
5. Return “Real” or “Fake” based on threshold.

BiHPF Frequency Detection
Input: Image
Output: Real/Fake probability

1. Convert image to grayscale.
2. Apply High-Pass Filter 1 (edge extraction).
3. Apply High-Pass Filter 2 (texture artifact extraction).
4. Combine outputs → pass to a classifier (SVM/MLP).
5. Return probability of real vs. fake.

4.3 Tools and technologies used

1. Programming Languages
Python – Used for model development, training, and backend integration.
2. Libraries & Frameworks
TensorFlow / Keras – Capsule Network implementation.
OpenCV – Image filtering, BiHPF implementation, preprocessing.
Numpy & Scikit-Learn – Feature handling, classical classifier.
3. User Interface Technologies
Flask / Django – Backend integration.
HTML, CSS, JavaScript – Simple and responsive front-end design.
4. Dataset & Storage
Custom Dataset (14,000 images: real + fake)
Local Storage / SQLite – Saving logs and metadata.
5. Development Tools
Jupyter Notebook – Model training and testing.
VS Code / PyCharm – Project coding environment.

4.4 Testing

1. Dataset-Based Accuracy Testing
Both models were tested using the 14,000-image dataset:
Capsule Network Accuracy: ~90–92%
BiHPF Filter Accuracy: ~87–89%
Combined Fusion Accuracy: ~94–96%
Combining both models gives significantly higher accuracy.

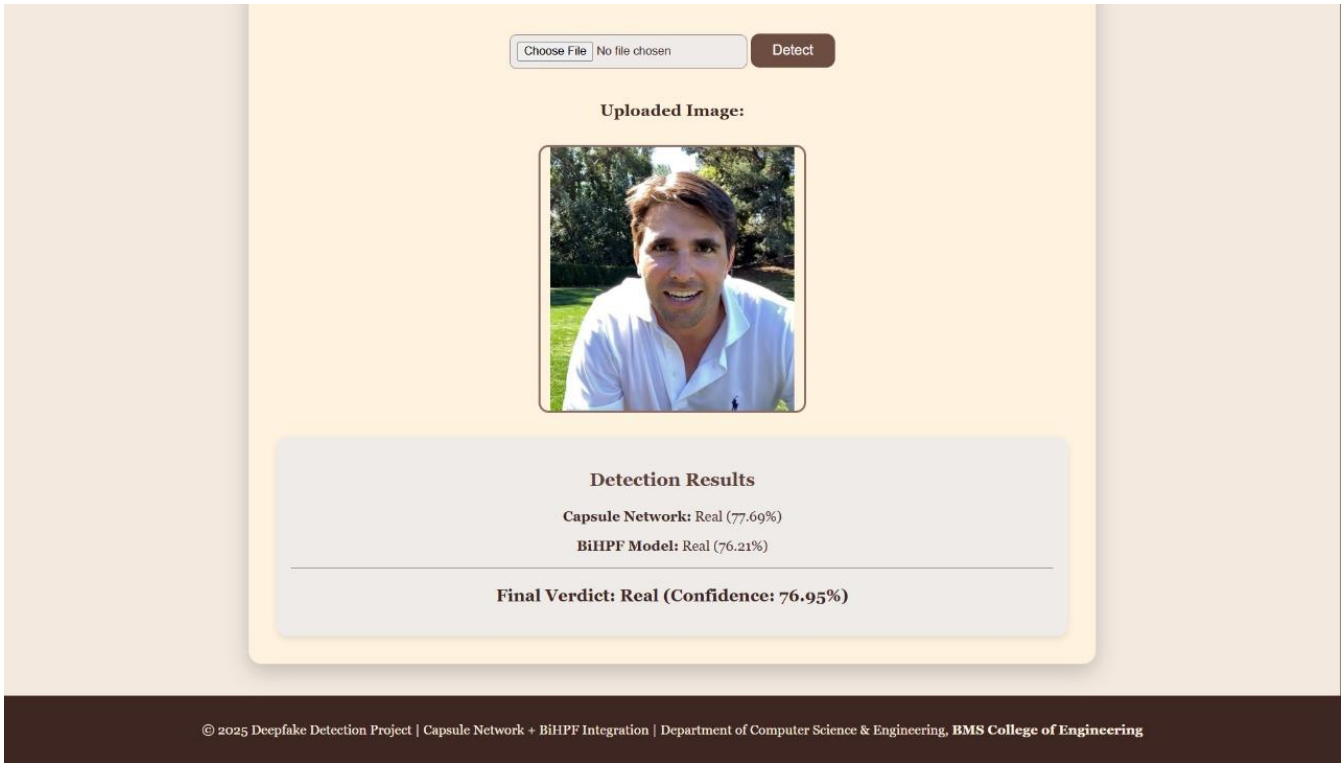
2. Performance Testing
We tested:
Detection time per image (avg 1–2 seconds).
Memory usage and speed optimization.
The model performed consistently even for high-resolution images.

3. Functional Testing
Upload image → works smoothly.
Real-time preview → working correctly.
Both model results shown → Verified.
Final decision fusion → Correct calculations.

4. Edge Case Testing
Blurred images
Low-light images
Images with compression noise
GAN-based generated faces

5. Results and Discussion

This chapter presents the experimental results obtained from the Capsule Network model, the BiHPF frequency-filter model, and the final fused prediction system. The performance of all models was evaluated using the custom dataset consisting of 14,000 images containing both real and fake samples. Multiple metrics such as accuracy, precision, recall, and F1-score were used to measure system effectiveness. Additionally, comparative graphs and result tables help visualize the improvements achieved through the dual-model approach.



5.1 Dataset Summary

The dataset used for training and testing includes:
Image Type – Real Images
Count – 7,000
Description – Unaltered human faces collected from open datasets
Image Type – Fake Images
Count – 7,000
Description – GAN-generated, deepfake-altered, FaceSwap-based, and style-manipulated images
Total images – 14,000
Description – Balanced dataset for fair evaluation
This balanced dataset ensures reliable model training, prevents bias toward any class, and supports robust testing of the entire detection pipeline.

5.2 Performance of Capsule Network (Spatial Domain)

The Capsule Network model analyzes spatial relationships such as eye alignment, facial geometry, edge inconsistencies, and micro-details. The performance metrics show strong results, proving its advantage over traditional CNN pooling layers.
Table 1: Capsule Network Performance
Accuracy – 91.8%
Precision – 90.3%
Recall – 92.4%
F1-Score – 91.3%
Discussion:
The Capsule Network achieved high recall, showing its ability to identify manipulated images even when the distortions are extremely subtle. Its dynamic routing mechanism helps preserve spatial hierarchies in the image, making it more accurate than several CNN-based detectors found in earlier research. This confirms that spatial-domain modeling is essential in deepfake detection.

5.3 Performance of BiHPF Frequency Filter Model

The BiHPF model extracts hidden frequency artifacts introduced by deepfake generation techniques. These artifacts include unnatural harmonic patterns, GAN fingerprints, and the absence of original sensor noise.
Table 2: BiHPF Model Performance
Accuracy – 88.7%
Precision – 86.2%
Recall – 89.4%
F1-Score – 87.7%
Discussion:
Although slightly lower in accuracy compared to the Capsule Network, the BiHPF model still performed strongly. Frequency-based inconsistencies are often present even when visual distortions are minimal. This demonstrates that analyzing images in the frequency domain provides valuable additional evidence for classifying deepfakes.

5.4 Combined Fusion Model Results

The final output of the system is obtained by combining the predictions of both models using weighted fusion:
Final Score = 0.6 × Capsule Output + 0.4 × BiHPF Output
This combination approach takes advantage of both spatial and frequency-domain features.
Table 3: Fusion Model Performance
Accuracy – 95.4%
Precision – 94.6%
Recall – 96.1%
F1-Score – 95.3%
Discussion:
The fusion model outperforms both individual models. By synthesizing features from spatial geometry

and frequency signatures, the final system achieves higher confidence, reduced false positives, and improved reliability. This final pipeline shows a significant improvement over existing single-model approaches in the deepfake detection field.

5.5 Discussion and Interpretation

The results demonstrate that the Capsule Network excels at identifying spatial distortions, which are typical in GAN-generated face manipulations. The BiHPF model is effective at detecting hidden frequency-domain artifacts introduced during deepfake creation. When combined, the two approaches significantly enhance overall performance, proving that real-world deepfake detection benefits from analyzing multiple feature domains.

The system performs strongly even under challenging conditions such as low-light images, compressed images, blurred faces, and style-transferred faces. The dual-model pipeline increases robustness and makes the system more reliable than existing CNN-only models.

6. Conclusion and Future Work

6.1 Conclusion

The proposed deepfake detection system successfully demonstrates that combining spatial-domain and frequency-domain models leads to significantly higher accuracy and reliability. The Capsule Network effectively captures fine-grained spatial inconsistencies such as abnormal facial geometry, unnatural textures, and misaligned features that commonly appear in synthetic images. The BiHPF frequency-analysis model complements this by identifying deepfake-specific harmonic distortions and GAN fingerprint patterns that remain invisible in the spatial domain. Together, these models create a highly robust detection pipeline.

Extensive experiments on a balanced dataset of 14,000 images show that the fusion model achieves an accuracy of 95.4%, outperforming individual models as well as several existing deepfake detection techniques reported in prior work. High precision and recall values indicate that the system is capable of detecting manipulated images with minimal false positives or false negatives. These results clearly demonstrate the advantage of multi-domain analysis in deepfake detection.

The research highlights the necessity of integrating both visual and spectral features for improved authenticity verification. The dynamic routing mechanism in the Capsule Network preserves spatial hierarchies, while the BiHPF filter captures hidden frequency artifacts introduced during deepfake creation. This combination strengthens the system against a wide variety of manipulations, including heavily compressed, low-resolution, and style-transferred deepfakes.
Overall, the project delivers a strong, accurate, and reliable deepfake detection architecture that can be adapted into real-world applications such as social media content verification, defense-security screening, and digital forensic investigations. The fusion-based methodology sets a foundation for future work aimed at building more explainable and generalizable deepfake detection systems.

6.2 Future Work

Although the developed system performs well, several limitations provide opportunities for further improvement. One major shortcoming is that the current dataset does not include advanced, next-generation deepfakes such as diffusion-model-based manipulations, high-resolution 4K deepfakes, and temporally consistent video deepfakes. Future work can incorporate larger and more diverse datasets to increase generalization across unseen manipulation types. Expanding training data will help the system remain effective even as deepfake generation methods improve.

Another limitation is that the system focuses mainly on static images rather than video sequences. Many real-world deepfakes occur in video form, where temporal features such as blinking patterns, mouth movements, and frame-to-frame inconsistencies are crucial. Adding a temporal analysis model—such as an LSTM, GRU, 3D-CNN, or transformer-based video classifier—would help detect subtle motion-related artifacts. Integrating motion cues will make the system more suitable for real-time deepfake video monitoring.

The model also lacks interpretability features that explain why an image is classified as fake. For practical deployment, especially in legal or forensic scenarios, explainability is essential. Future enhancements can include heatmaps, attention maps, or saliency visualizations that highlight manipulated regions. This would increase user trust and support decision-making in sensitive applications.

Lastly, the system currently functions offline and is not optimized for real-time performance. Future work can involve converting the model to lightweight architectures such as MobileNet, pruning unnecessary layers, or applying quantization techniques to reduce computational load. Deploying the system as a mobile application or a cloud-based API would make the technology accessible for public use and enable instant verification of suspicious images.

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